

# Multiplying and Dividing in Scientific Notation

*Teacher's Edition*

## The Period

**MA.8.NSO.1.5** Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

**Must** Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

**Must not** No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

**Target** I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

**MTR** **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 6

| Segment                             | Slides | Time          |
|-------------------------------------|--------|---------------|
| Settle in; post the learning target | 1      | 1 min         |
| Warm-Up                             | 2-3    | 5 min         |
| Notes                               | 4-6    | 11 min        |
| Example 1                           | 7-8    | 3 min         |
| Your Turn                           | 9-10   | 3 min         |
| Example 2                           | 11-12  | 3 min         |
| Your Turn                           | 13-14  | 3 min         |
| Whiteboards                         | 15-32  | 13 min        |
| Independent Set                     | 33     | 6 min         |
| IXL                                 | 34     | 5 min         |
| <b>Total</b>                        |        | <b>53 min</b> |

### Say these three things out loud today

1. The powers of ten never have to match when you multiply or divide. They can be as far apart as they like.
2. Multiply the fronts and add the backs; divide the fronts and subtract the backs, top minus bottom.
3. A product of two coefficients under 10 is very often over 10. Look at the coefficient before you put the board up.

## Slide by Slide

---

1. **Multiplying and Dividing in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q1 and Q2 are the two halves of today, side by side.
3. **Warm-Up — Answers.** Reveal. 1.  $6.2 \times 10^6$  2.  $10^5$  3. 13 4. 3
4. **Notes — Copy both lines.** Multiply the fronts, add the backs. The powers of ten never have to match when you multiply.
5. **Notes.** Top minus bottom, always in that order. And subtracting a negative goes up.
6. **Notes.** Coefficient over 10, point left and exponent up. Coefficient under 1, point right and exponent down. Same repair as yesterday.
7. **Example 1 — Multiplying, and the answer needs finishing.** SPOKEN ONLY.
8. **Example 1 — Worked.** Regroup, multiply, add the exponents, then repair.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the other sign on the small exponent.
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — Multiplication and division in one expression.** SPOKEN ONLY.
12. **Example 2 — Worked.** The intermediate value is  $60 \times 10^6$ , which is exactly the unfinished form — and it does not have to be repaired yet, because the division is still to come.
13. **Your Turn — Same steps, your numbers. Boards up when done.** A negative exponent inside the product and a division that pushes the answer below 1.
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.**  $(3 \times 10^4)(2 \times 10^5) \rightarrow 6 \times 10^9$   $6 \times 10^{20}$  — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception:  $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$ ];  $5 \times 10^9$  — the coefficients were added [8.NSO.1.5: the operation is multiplication]
- 17-18. **Board 2.** A printer makes  $4 \times 10^3$  copies per hour. It runs for  $2.5 \times 10^2$  hours. Paper costs 3 cents a sheet. How many copies does it make?  $\rightarrow 1 \times 10^6$   
The paper cost of 3 cents a sheet is not needed. anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost];  $10 \times 10^5$  — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]
- 19-20. **Board 3.**  $\frac{9.6 \times 10^8}{1.2 \times 10^3} \rightarrow 8 \times 10^5$   $8 \times 10^{11}$  — the exponents were added [8.NSO.1.5: division subtracts the exponents];  $8.4 \times 10^5$  — the coefficients were subtracted instead of divided [8.NSO.1.5]
- 21-22. **Board 4.**  $\frac{2.65 \times 10^{11}}{5 \times 10^{13}} \rightarrow 5.3 \times 10^{-3}$   $0.53 \times 10^{-2}$  — right value, but 0.53 is not at least 1 [8.NSO.1.5: the coefficient satisfies  $1 \leq a < 10$ ];  $5.3 \times 10^{-1}$  — the point moved but the exponent did not
- 23-24. **Board 5.**  $(2 \times 10^4)(3 \times 10^5)$ . Which is the product?  $\rightarrow C - 6 \times 10^9$  (A) the exponents were subtracted,  $4 - 5$  [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception:  $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$ ]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is

multiplication] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**25-26. Board 6.**  $(5 \times 10^6)(4 \times 10^3)$ . Which is the product, written in scientific notation? → **D —  $2 \times 10^{10}$**  (A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving  $12 \times 10^9$  instead of  $1.2 \times 10^{10}$ ]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**27-28. Board 7.**  $\frac{3 \times 10^8}{3 \times 10^{-4}}$  →  **$1 \times 10^{12}$**   $1 \times 10^4$  — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds];  $1 \times 10^{-12}$  — bottom minus top [8.NSO.1.5: top minus bottom]

**29-30. Board 8.**  $(8.4 \times 10^7) \div (n \times 10^2) = 4.2 \times 10^5$ . What is the value of n? →  **$n = 2$**   $4.2$  — the coefficients were subtracted,  $8.4 - 4.2$  [8.NSO.1.5: the coefficients divide];  $0.5 - 4.2 \div 8.4$  instead of  $8.4 \div 4.2$

**31-32. Board 9.** For  $(7.2 \times 10^9) \div (2.4 \times 10^4)$ , Marcus writes  $4.8 \times 10^5$  and Sara writes  $3 \times 10^{13}$ . Name what each one did wrong, and give the correct quotient. →  **$3 \times 10^5$**  'they both got it wrong' — neither error is named [the item asks what each one did];  $4.8 \times 10^5$  — Marcus's error repeated

**33. Independent Set — Six questions. On your own, in silence.** Hand out the Independent Set.

**34. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

## Misconceptions to Watch

---

**Board 1.**  $6 \times 10^{20}$  — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception:  $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$ ]

**Board 2.** anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]

**Board 3.**  $8 \times 10^{11}$  — the exponents were added [8.NSO.1.5: division subtracts the exponents]

**Board 4.**  $0.53 \times 10^{-2}$  — right value, but 0.53 is not at least 1 [8.NSO.1.5: the coefficient satisfies  $1 \leq a < 10$ ]

**Board 5.** the exponents were subtracted,  $4 - 5$  [8.NSO.1.5: multiplication adds the exponents]

**Board 6.** the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]

**Board 7.**  $1 \times 10^4$  — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]

**Board 8.**  $4.2$  — the coefficients were subtracted,  $8.4 - 4.2$  [8.NSO.1.5: the coefficients divide]

**Board 9.** 'they both got it wrong' — neither error is named [the item asks what each one did]

**Materials** Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether  $6 \times 10^9$  and  $6 \times 10^{20}$  could possibly both be the answer to whiteboard 5.